

AquaValley — Project Plan

Water Management System · New Valley Governorate, Egypt

14 Development Days + 3 Demo/Presentation Days · 2 Developers

14

Development Days

3

Demo Prep Days

2

Developers

21/21

Sequences Covered

Dev A

Backend · API · DB ·
Deploy

Dev B

Frontend · UI · Charts
· UX

Shared

Sync · Integration ·
Kickoff

PHASE 1 · FOUNDATION & AUTH & CORE API — Days 1–7

Sun · Day 1	Mon · Day 2	Tue · Day 3	Wed · Day 4
Both · Shared Project Kickoff (2 hrs) GitHub · branches · API contracts doc · .env.example	Dev A · Backend / API Auth.js v5 + RBAC JWT · GOV_ADMIN/FARMER · bcrypt · middleware · SQ-01 [Auth] [■ BLOCKER]	Dev A · Backend / API Wells CRUD API GET/POST/PUT · Zod validation · audit_log transaction · SQ-04,05 [API]	Dev A · Backend / API Sensor Ingest + Threshold Engine POST /api/ingest · validate · evaluate rules · create alerts · SQ-21,06 [Core Logic] [■ Critical]
Dev A · Backend / API Database Schema Prisma + MongoDB · users, wells, readings, alerts, irrigation, quota, audit_logs [Prisma] [MongoDB]	Dev B · Frontend / UI Design System CSS vars · color tokens · Typography · Button/Badge/Card [DS]	Dev B · Frontend / UI Login Page Form · error states · role redirect · SQ-01 [Auth UI]	Dev B · Frontend / UI Gov Dashboard Page 4 KPI cards · RSC streaming · Leaflet map · well markers · SQ-02 [Dashboard] [Leaflet]

Dev B · Frontend / UI Next.js 15 Init App Router · Tailwind · shadcn/ui · Cairo font · RTL layout shells [Next.js 15] [RTL]	Dev B · Frontend / UI Sidebar + Topbar Gov nav · Farm nav · active states · mobile collapse [Layout]	Dev B · Frontend / UI Seed Data Script 5 districts · 20 wells · real GPS · crop profiles · Faker.js [Data]	
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Thu · Day 5	Fri · Day 6	Sat · Day 7
Dev A · Backend / API Alerts API + Notification Alert lifecycle · Resend email · SQ-06,07 [Alerts] [Resend]	Dev A · Backend / API Quota System API monthly_summaries · hard block at 100% · SQ-20 [Business Logic]	Both · Shared Buffer / Catch-up Day Any overflow from Days 1–6
Dev B · Frontend / UI Well Detail Page Radial SVG gauge · 30-day Recharts · 30s live poll · SQ-03 [Charts]	Dev B · Frontend / UI Alerts Management Page Filterable table · ack/resolve · status lifecycle · SQ-07 [Gov Portal]	Both · Shared Integration Checkpoint #1 Login → Dashboard → Well Detail — first real full flow, no mocks
Sync · EOD EOD Sync (30 min) Review API contracts for Week 2	Dev B · Frontend / UI Wells Registry Page Full table · district filter · register well modal · SQ-04 [Gov Portal]	

PHASE 2 · ADVANCED FEATURES + AI INTEGRATION — Days 8–14

Sun · Day 8	Mon · Day 9	Tue · Day 10	Wed · Day 11
Dev A · Backend / API Claude AI Irrigation API Gather context · call Claude · Zod JSON output · quota-aware · SQ-16 [Claude API] [■ Critical]	Dev A · Backend / API Irrigation Trigger + IoT Mock Pipeline server action · quota check · mock webhook · SQ-17 [IoT Mock]	Dev A · Backend / API Aquifer Forecast Engine Linear regression · 5yr/10yr/25yr · risk flags · SQ-13 [Analytics]	Dev A · Backend / API Report Generator @react-pdf/renderer · CSV · 3 report types · SQ-10 [PDF] [CSV]
Dev B · Frontend / UI Farm Dashboard KPI cards · quota bar · AI rec card · 60s live poll · SQ-14 [Farm Portal]	Dev B · Frontend / UI AI Irrigation Plan Page Loading skeleton · plan card · zone breakdown · SQ-16 [AI UI]	Dev A · Backend / API Valve Control API District auth · PENDING state · IoT command · audit · SQ-11 [Safety-Critical]	Dev A · Backend / API Vercel Cron Job *15 cron · batch 50/parallel · /api/ingest?manual=true · SQ-21 [Vercel] [Cron]
	Dev B · Frontend / UI Irrigation Live Monitor Zone status · liter counter · 30s polling · SQ-17 [Real-time]	Dev B · Frontend / UI Water Distribution Page Bar charts by type+district · 12-month trend · SQ-08 [Charts]	Dev B · Frontend / UI Forecast Page + Report UI Projection chart · scenario toggle · date range picker · SQ-13,10 [Gov Portal]
			Dev B · Frontend / UI Valve Modal Safety confirmation · command form · SQ-11 [Gov Portal]
Thu · Day 12	Fri · Day 13	Sat · Day 14	
Dev A · Backend / API User Management API Create user · temp password · email welcome · farm scoping · SQ-12 [RBAC]	Both · Shared Integration Day Wire every UI to real APIs · fix broken flows · full end-to-end run	Both · Shared Buffer Day Overflow from Day 13	
Dev B · Frontend / UI Remaining Farm Pages Crop Profile · Soil Readings · History · Quota · SQ-15,18,19,20 [Farm Portal]	Dev A · Backend / API Deploy to Vercel MongoDB Atlas · Resend · OpenMeteo · Claude API keys · domain	Both · Shared Smoke Test Every page · every flow · no crashes · LGTM	

Dev B · Frontend / UI User Mgmt + Settings Pages User table · create modal · threshold forms · SQ-09,12 [Gov Portal]		
Sync · EOD EOD Sync (1 hr) Full 21/21 SQ check — what is done, what remains		

PHASE 3 · POLISH, QA & DEMO PREPARATION — Days 15–17

Sun · Day 15	Mon · Day 16	Tue · Day 17 — DEMO
Dev A · Backend / API E2E Tests + Bug Fixes Playwright smoke tests · security headers · rate limiting [Testing] [Security]	Dev A · Backend / API Performance Audit Lighthouse · DB indexes on readings+alerts · caching headers [Performance]	Both · Shared Demo Rehearsal x3 (morning) 3-minute story · each dev presents once · time Q&A;
Dev B · Frontend / UI Landing Page Problem in one sentence · portal screenshots · CTA · mobile ■ Judge Impact	Dev A · Backend / API Demo Data Setup Seed realistic data · manual trigger ready · alert prepared [Demo Prep]	Dev A · Backend / API Final Deploy Check Production URL · env vars · cron running · API healthy
Dev B · Frontend / UI Loading Skeletons Skeleton for every page · empty states · error boundaries [UX Polish]	Dev B · Frontend / UI Mobile Responsive Farm portal on phone · breakpoints · touch buttons [Mobile]	Dev B · Frontend / UI Slide Deck Problem · Solution · Tech · Features · Live Demo · Roadmap
	Dev B · Frontend / UI Micro-interactions Toast system · transition animations · form feedback [UX]	Both · Shared Hackathon Presentation LIVE DEMO · judges · win ■

Key Milestones

Day 2	Auth + Login Working	JWT issued · GOV_ADMIN/FARMER redirect works · first real login
Day 5	Gov Portal V1 Live	Dashboard + Map + Well Detail + Alerts — all with real DB data
Day 7	Integration Checkpoint	First full flow: Login → Dashboard → Well → Alert — no mocks
Day 9	AI Feature Live	Claude API generates real irrigation plans based on ET0 + quota + soil
Day 12	Feature Complete — 21/21 SQ	Every sequence diagram from UML is implemented and working
Day 14	Deployed + Stable	Production URL live · integration complete · no crashes
Day 16	Demo Ready	Landing page live · skeletons · mobile polish · data seeded
Day 17	Hackathon Presentation	Demo rehearsed x3 · slide deck · manual trigger ready

Responsibilities Split

Dev A — Owns	Dev B — Owns
• All Server Actions + API Routes	• All React Server Components (App Router)
• Prisma schema + MongoDB Atlas setup	• Arabic RTL design system + Cairo font
• Auth.js v5 + RBAC + Middleware	• Leaflet map integration (dynamic import, SSR:false)
• Sensor ingest + Threshold engine (SQ-21, 06)	• Recharts area/bar charts + custom SVG gauges
• Claude AI Irrigation API — main feature (SQ-16)	• TanStack Query live polling (30s / 60s)
• Quota enforcement + hard block at 100%	• Gov Portal — 9 pages complete
• Valve control + audit trail (SQ-11)	• Farm Portal — 6 pages complete
• PDF/CSV report generator (SQ-10)	• Loading skeletons + empty states + error boundaries
• Aquifer forecast linear regression (SQ-13)	• Landing page (judges see this first)
• Vercel Cron job + deployment + env config	• Mobile responsive — Farm Portal
• Performance audit + DB indexes	• Micro-interactions + Toast notification system
• Demo data seeding + /api/ingest?manual=true	• Slide deck for presentation day

Tech Stack

Layer	Technology	Notes
Framework	Next.js 15	App Router + RSC + Server Actions + Streaming
Language	TypeScript	Strict mode · shared types between Dev A and Dev B
Styling	Tailwind CSS + shadcn/ui	Arabic RTL · Cairo font · custom design system
Database	MongoDB Atlas	Prisma ORM · transactions · audit_logs collection
Auth	Auth.js v5	JWT · GOV_ADMIN / FARMER RBAC · HttpOnly cookie
State / Data	TanStack Query	Live polling 30s/60s · cache invalidation · RSC hydration
Charts	Recharts + SVG	Area charts · bar charts · custom radial gauges
Map	Leaflet.js	Dynamic import SSR:false · real GPS coords per district
AI	Anthropic Claude API	claude-sonnet-4-6 · structured JSON output · quota-aware prompt
Weather	Open-Meteo API	ET0 · temperature · 3-day forecast · free, no key
Email	Resend	Critical alert notifications · welcome emails
PDF/CSV	@react-pdf/renderer	Report export · serverless-friendly · no Puppeteer
Deployment	Vercel	Serverless · Cron jobs (*15) · Edge config
CI/CD	GitHub Actions	PR checks · type check · lint before merge

The Golden Rule — API Contract

Dev A defines TypeScript types for every API response on Day 1. Dev B works against these types using mock data. When Dev A finishes the real implementation, Dev B switches with zero changes.

Example shared type (defined by Dev A, used by Dev B):

```
export type WellDTO = { id: string district: '████████' | '████████' | '████████' |
'████████' | '████' level: number // 0-100 percent flow: number // m3/h status: 'active' | 'low' |
'critical' | 'maintenance' lat: number lng: number } export type AlertDTO = { id: string type:
'CRITICAL' | 'WARNING' | 'INFO' status: 'OPEN' | 'IN_PROGRESS' | 'RESOLVED' wellId: string
message: string createdAt: Date }
```

Never share code directly — share types. Dev A can rewrite the entire database schema as long as the DTO types stay the same. Dev B can restyle every component as long as it consumes the same types. This is what lets two people work in parallel without stepping on each other.

Risks & Mitigations

HIGH	Claude API latency (8+ seconds for irrigation plan)	Mitigation: Streaming response + skeleton animation + cache plan for 30 min
HIGH	Dev B blocked waiting for Auth API from Dev A (Day 2)	Mitigation: Dev A exports TypeScript types + mock session first 2 hours. Dev B works on Design System in parallel
MEDIUM	No real IoT — how to show live alerts in demo?	Mitigation: /api/ingest?manual=true sends reading at 1.35x baseline → fires critical leakage alert instantly
MEDIUM	Leaflet SSR crash in Next.js App Router	Mitigation: dynamic() => import('./Map'), { ssr: false }) — documented pattern, 5 minute fix
LOW	Integration Day (Day 13) reveals major bugs	Mitigation: Day 7 checkpoint + Day 14 buffer absorb surprises. Early integration = early discovery
LOW	Arabic text layout issues in charts/tables	Mitigation: Recharts supports RTL natively. Custom SVG gauges built by Dev B from scratch

Demo Story — 3 Minutes

This is the exact sequence to rehearse on Day 17 morning (x3 runs). It covers 15 of the 21 sequence diagrams in under 3 minutes.

1	Open Landing Page	10 sec	Judges see the problem in one sentence before login
2	Login as Gov Admin	15 sec	JWT issued → auto-redirect to Gov Dashboard
3	Dashboard shows leakage	20 sec	Critical alert on Kharga well visible immediately
4	Trigger manual ingest	5 sec	/api/ingest?manual=true → live alert fires (IoT simulation)

5	Acknowledge the alert	20 sec	OPEN → IN_PROGRESS → demonstrate audit trail
6	Well Detail page	25 sec	Radial gauge + 30-day chart + close the valve (confirmation)
7	Aquifer Forecast page	20 sec	Show Kharga hitting critical depth by 2028
8	Switch to Farm Portal	10 sec	Same session — different portal, different UI
9	Request AI Irrigation Plan	30 sec	Loading skeleton → Claude generates zone-by-zone plan
10	Activate irrigation	20 sec	Live liter counter runs → completion toast
11	Q&A;	45 sec	Judges ask questions — point to UML diagrams if needed